

CYBER-TANK ARENA: NOMADS

An Interactive 3D Browser Game Built with Three.js

Interactive Graphics Course • Final Project • 2025



01

General Overview

A futuristic combat arena with procedurally generated visuals

Procedural Textures

All visuals generated in JS
No external image assets

Wave-Based Combat

Survive increasingly
difficult enemy waves

Full 3D Scene

Hierarchical models, PBR
materials, dynamic lighting

02

Architecture

Modular file structure — each file handles one concern

main.js

Game loop, state, collision

arena.js

Ground, obstacles, starfield

scene.js

Renderer, camera, fog

lights.js

4 light types, headlight

tank.js

Player tank hierarchy

textures.js

5 procedural Canvas textures

enemy.js

Drone AI, tiers, death anim

input.js

Keyboard, mouse raycasting

bullet.js

Player & enemy projectiles

hud.js

Score, HP bar, game-over

03

Hierarchical Tank Model

5-level scene graph with independent animations

tankGroup

Root — global position & rotation

chassis

Body — movement direction, suspension wobble

wheels (6)

Spin proportional to speed

turretGroup

Aims independently toward mouse cursor

barrelGroup

Recoil animation on each shot (Tween.js)

muzzleTip

Bullet spawn point + muzzle flash anchor

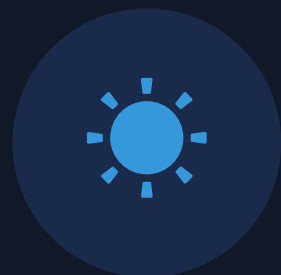
Key Properties

- PBR material: metalness 0.7, roughness 0.6
- Emissive glow: intensity 0.4 (orange)
- Suspension: roll on turn, tilt on accel
- Turret: smooth interpolation toward mouse
- Barrel: Tween.js recoil on every shot

04

Lighting System

4 light sources working together for atmospheric visuals



AmbientLight

Blue-tinted base illumination
Intensity 1.8 • Shifts hue per wave



DirectionalLight

Sun / moon • Primary shadow caster
4096×4096 shadow map



SpotLight

Tank headlights •
Toggleable (L key)
Penumbra 0.4 • Visible cone mesh



PointLight

Muzzle flash + enemy death flash
Sprite glow with additive blending

05

Procedural Textures

5 Canvas-generated textures — zero external image files



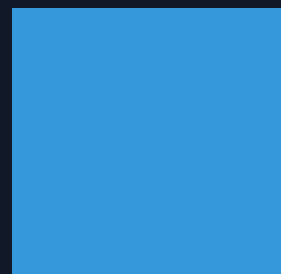
Base Colour

Dark metal with scratches, dents, orange accents



Emissive Map

Circuit-board traces, junction nodes, glow dots



Normal Map

Rivets, armour plates, surface irregularities



Ground (512×512)

Dirt speckles, tyre tracks, rocks, orange grid



Roughness Map

Noise base, directional brush marks, wear spots

All generated via HTML Canvas API in textures.js

06

User Interaction

Keyboard, mouse, and UI controls

WASD / Arrows

Tank movement with smooth acceleration / deceleration

Mouse

Turret aiming via raycaster (ground-plane intersection)

Left Click

Fire projectile • 250 ms cooldown
Barrel recoil + muzzle flash

L Key

Toggle headlights on/off
HUD indicator updates

X Key

Toggle wireframe on all scene meshes

Tab Key

Config panel (pauses game)
Colours, camera, difficulty

07

Animations

10+ distinct animation types using JS + Tween.js

Tank

- Smooth acceleration / deceleration
- Chassis rotation toward movement
- Wheel spin proportional to speed
- Turret tracking toward mouse
- Barrel recoil (Tween.js)
- Suspension wobble (roll + tilt)
- Invulnerability flash

Enemies

- Pursuit movement + tracking
- Rotor spin (8 rad/s)
- Core pulse + rotation
- Hover bob (sine wave)
- Hit flash (0.15 s)
- Death: light flash + shrink

Bullets & Arena

- Trail lines (5-pt / 3-pt buffer)
- Glow sprites (additive blend)
- Starfield twinkling + rotation

08 Visual Effects & Rendering

Polish that elevates the game beyond basic Three.js

Tone Mapping

Cineon at exposure 1.2
Filmic colour reproduction

Fog

FogExp2, density 0.018
Seamless horizon blending

Shadow Maps

PCFSoftShadowMap
Soft blurred shadow edges

Muzzle Flash

Additive sprite + PointLight
80 ms Tween.js fade

Enemy Death

PointLight flash 6→0
Scale shrink with easing

Low HP Vignette

Red border + box-shadow
HP bar pulsing animation

Bullet Trails

Line geometry, manual
vertex
update each frame

Bullet Glow

Additive-blended sprites
Procedural radial gradient

09

Game Mechanics

Wave progression, scoring, and difficulty scaling

Wave System

- Enemies per wave: $5 + \text{wave} \times 2$
- Staggered spawn: 400 ms apart
- Speed scales with wave number
- 3 visual tiers (red → purple → chrome)
- Difficulty: Easy / Normal / Hard

100

Starting HP

8

Bullet Damage

15/s

Contact Damage

0.5s

Invuln. Window

Score: $100 + \text{wave} \times 10$ • XP: 25 per kill

10

Collision & Camera

Custom physics and three camera modes

Collision Detection

- Circle-vs-AABB intersection
- Squared-distance (no sqrt)
- Tank vs obstacles + boundaries
- Player bullets vs enemies
- Enemy bullets vs player
- Enemy obstacle avoidance

Default

Top-down follow
Smooth interpolation

Chase

Close behind tank
Follows movement dir.

Orbit

Circles at radius 10
Continuous rotation



Course Requirements

Fulfilment of Interactive Graphics course objectives



Hierarchical Modelling

5-level tank scene graph
Independent animations at each level



Lighting

4 light types: Ambient, Directional,
SpotLight, PointLight



Textures

5 procedural maps: base colour,
normal, roughness, emissive, ground



User Interaction

WASD, mouse aiming, shooting,
light toggle, config panel, restart



Animations

10+ types: tank, enemy, bullet,
camera, wave transitions, effects



Rendering

PBR materials, shadows, tone
mapping,
fog, additive blending, trails

Thank You

Cyber-Tank Arena: Nomads

A fully playable interactive 3D game demonstrating:

Hierarchical modelling • Dynamic lighting • Procedural textures
Wave-based combat • Collision detection • Full restart flow

Built entirely in JavaScript with Three.js — no external assets.

